

JAMES HART

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[LinkedIn](#) | [GitHub Repos](#) | [GitHub Blog](#) | [GitHub Resume](#)

Senior software engineer with 10+ years of professional experience in full-stack development and DevOps practices in Agile environments across finance and other industries. Proven track record in building a wide range of scalable, high-performance applications and improving on existing system architecture. Extensive automation experience. Strong background implementing CI/CD pipelines and secure Cloud hosted environments to streamline deployment of microservices and multi-tier applications. Specializing in Python, .NET, and JavaScript.

SKILLS

Languages: Python | .NET (C#/F#) | JavaScript | Java | PowerShell | Go | PL/SQL | T-SQL

Development Frameworks: ASP.NET Core | Python Flask | JavaScript: React, Node.js, Bootstrap, Angular | Spring Boot

Databases: SQL: Microsoft SQL Server, Oracle, PostgreSQL | NoSQL: Redis, MongoDB

Development Methodologies: Agile | Domain Driven Design (DDD) | Test Driven Design (TDD)

DevOps & CI/CD Tools: Azure DevOps Services | Jenkins | GitHub Actions | Docker | Kubernetes | Git | TFS | Bitbucket | Octopus Deploy | TeamCity | Jira

Cloud Platforms & Services: Microsoft Azure | Okta | Heroku

Configuration Management: Chef | Ansible | ARM Templates

Application Monitoring: Jaeger | OpenTelemetry | ITRS | New Relic | PRTG

Messaging Systems and Protocols: AMPS | Redis Pub/Sub | RabbitMQ | RESTful APIs | gRPC

HONORS & ACHIEVEMENTS

- Designed and implemented a market data validation service to automate extraction, validation, and notification of invalid price curves, including robust error handling, configurable escalation scheduling, and integration with multiple notifications channels for timely issue escalation and reporting
- Developed Pester PowerShell scripts adopted by development teams company-wide to automate validation of application configurations during software upgrades, reducing manual effort by 99% (from 2 days to 30 seconds) and significantly minimizing troubleshooting time
- Automated the creation and configuration of Hyper-V Windows VMs using PowerShell and Chef, reducing setup time by 98% (from 2-4 days to 30 minutes)
- Automated software deployment processes using PowerShell including inserting XML data into databases and IIS configuration, decreasing setup time by 98% (4 hours to 5 minutes)
- Recognized twice at the company level for exceptional impact as a senior member of a small team, significantly improving company-wide development processes and setting new standards for efficiency and collaboration
- Recognized for excellence in managing complex release processes by ensuring comprehensive and compliant documentation that met external regulatory requirements, streamlining workflows, and enhancing team efficiency
- Featured interviewee on Pluralsight's [All Hands on Tech Podcast Episode 056](#) – Upskilling Tips and Tricks with Developer James Hart

EXPERIENCE

Software Engineer III

Bank of America – Houston, TX

02/2022 – Present

- Architect and develop scalable and resilient microservice-based systems for commodities market data management contributing to continual, iterative improvements for Oil, Natural Gas, and Emissions trading desks
- Facilitate ticking market data flow from Commodity Exchanges (e.g. NYMEX) to Front Office traders
- Report on feature development process statuses and developments for curve marking features that eliminate dependency on spreadsheets, improving data integrity and user efficiency
- Offer advanced technical support to key trading desk stakeholders for their successful transition to a robust price curve marking application, resulting in increased efficiency and data accuracy
- Develop visualizations and reporting for real-time data tracking through downstream systems, increasing transparency in service of continuous improvement of the curve marking application
- Mentor and guide junior developers in best practices for modern software engineering, fostering a culture of continuous learning and improvement
- Tools:** Python (Quartz), React/Redux, AMPS Messaging, BofA NoSQL DB, ITRS, Bitbucket, Jenkins

Software Engineer II

Bank of America – Houston, TX

10/2019 – 01/2022

- Lead the analysis of requirements and feature development necessary to retire a legacy application, coordinating with cross-functional teams and ensuring seamless collaboration
- Enhanced middle-office reconciliation applications, reducing data entry time from weeks to minutes and improving operational efficiency
- Debugged and maintained an Exchange limit monitoring application, ensuring optimal performance and reliability to end users
- Developed an ETL feed mapping Exchange accounts to trader books, enabling accurate position tracking and compliance with regulatory limits
- Tools:** ASP.NET/C#, MSSQL, Python (Quartz), BofA NoSQL DB, Bitbucket, Jenkins, Ansible

Senior Software Consultant

Credera – Houston, TX

05/2019 – 10/2019

- Collaborated with clients to solve their biggest technological challenges and contributed to a culture of continuous learning and improvement
- Supported a large-scale Microsoft Dynamics CRM installation and surrounding custom applications, ensuring system stability and performance
- Contributed to a DevOps community of practice and led events exploring new and emerging technologies, concepts, best practices and enhancing team skills in CI/CD processes for junior members
- **Tools:** PowerShell, MSSQL, Jenkins, Octopus Deploy, New Relic, PRTG

Senior Software Operations Engineer

Quorum Business Solutions Inc. – Houston, TX

03/2018 – 05/2019

- Managed operational processes to decrease development life cycle timelines under an agile software development methodology
- Developed tools, built scripts, and created documentation used for client-side web server and .NET framework based software installations
- Automated Azure DevOps release pipeline creation for deployment of software applications, increasing process efficiency by 90%
- Migrated n-tier application environments to Azure Cloud, allowing the business to evaluate cost and time savings
- Developed a web application to automate server information management, significantly improving software deployment processes
- **Tools:** PowerShell, .NET/C#, RabbitMQ, Oracle, MSSQL, Azure DevOps Services, Chef, Ansible, Azure, TeamCity

Senior DevOps Engineer

Quorum Business Solutions Inc. – Houston, TX

01/2017 – 02/2018

- Automated and standardized internal software deployment processes, reducing completion times from days to minutes
- Drove the adoption of best practices in software development and automation, minimizing manual errors and improving process reliability
- Designed and implemented a standardized SQL script template for managing database settings, streamlining database administration across environments and data refreshes
- Deployed and debugged application and database installations in clients' non-Production and Production environments
- **Tools:** PowerShell, .NET/C#, Oracle, MSSQL, Azure DevOps Services, TeamCity

Senior Software Implementation Consultant

Quorum Business Solutions Inc. – Houston, TX

04/2014 – 12/2016

- Implemented Quorum Land Suite software upgrade for major clients, leading small teams across multiple successful project completions
- Provided robust time and effort estimations throughout the project lifecycle from internal development through transition to long-term maintenance
- Developed an Oracle database package to streamline legacy data migration, reducing upgrade timelines from months to weeks
- **Tools:** Oracle, .NET/C#

Software Implementation Consultant

PowerPlan Inc. – Atlanta, GA

06/2012 – 03/2014

- Implemented PowerTax software for major corporate clients, providing expertise in utility tax accounting and Oracle SQL scripting
- Coordinated script migrations and database management with clients' IT
- **Tools:** Oracle SQL

PROJECTS & INTERESTS

MusicMan

- A music data management app demonstrating full stack integration ([my repo](#))
- **Technologies:** .NET C#, TypeScript/React, PostgreSQL, Docker

InvoiceWiz

- A microservice-based invoice generator with OpenTelemetry tracing ([my repo](#))
- **Technologies:** .NET C#, Vue.js, MongoDB, RabbitMQ, Azure Blob Storage, docker, Kubernetes (k8s), Jaeger, OpenTelemetry, gRPC

MediatR vs Discriminated Unions

- A comparison of the C# MediatR library with F# Discriminated Unions ([my repo](#) / [my write-up: mediatr-vs-discriminated-unions](#))
- **Technologies:** .NET C#/F#, MediatR

TicketPusher

- Ticket management website inspired by Jira utilizing GitHub Actions to deploy as a multi-container Azure App Service via an ACR ([my repo](#))
- **Technologies:** C#, Blazor, .NET Core Web API, EF Core, Azure SQL Database, Docker Compose, Azure Container Registry (ACR), Okta

EDUCATION

Bachelor of Arts in Mathematics, Minor in Computational and Applied Math

Rice University – Houston, Texas

08/2008 – 05/2012

- GPA: 3.8, Cum Laude
- Publication: [Energy-minimizing unit vector fields](#)
- [National Hispanic Merit Scholar](#)

Prior experience references available upon request.

